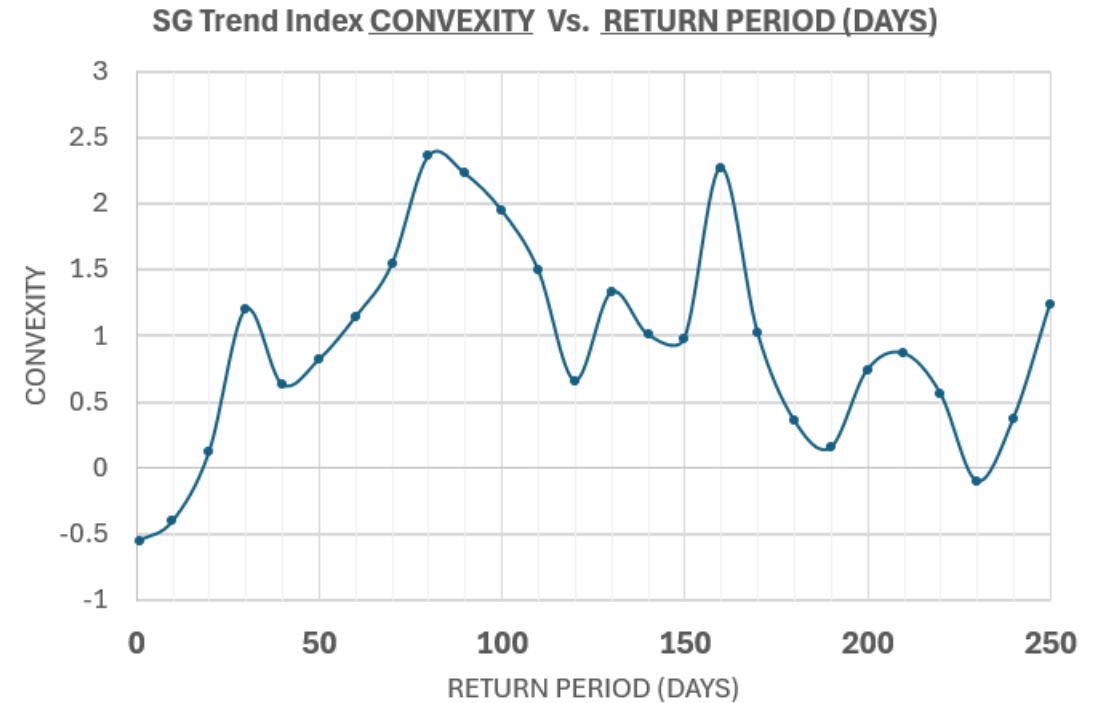
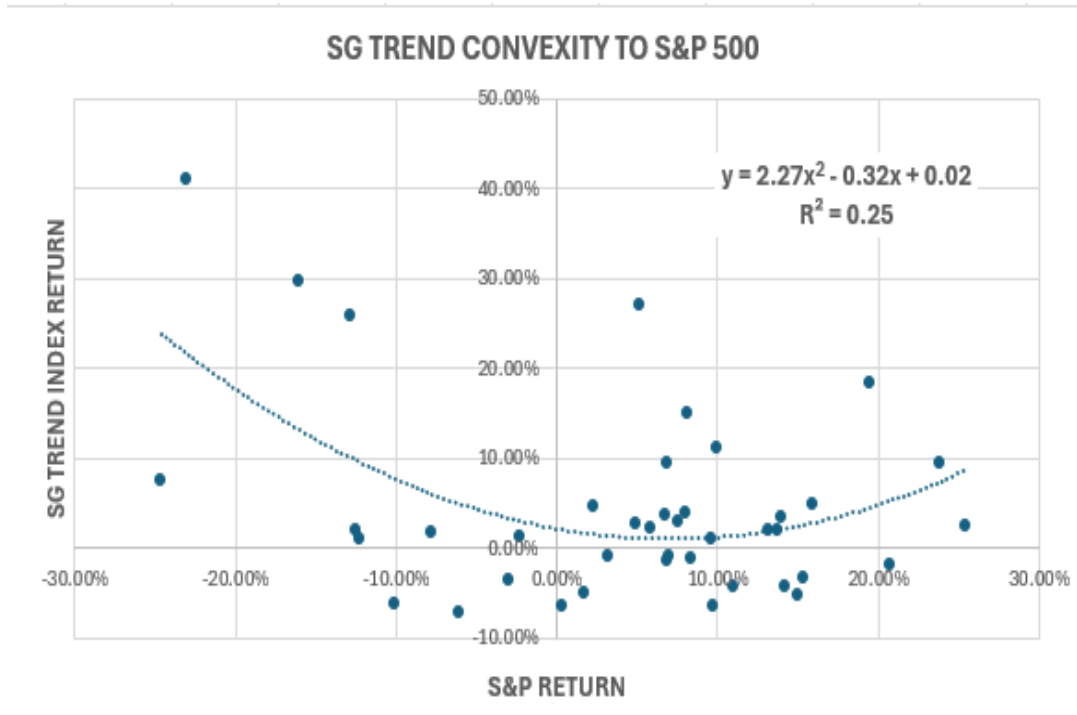


SG TREND INDEX CONVEXITY VS S&P 500

Define convexity as the co-efficient of the x^2 term in a quadratic line-of-best fit.

Then examine how the convexity term varies as the return-period is changed (from 1 to 250 days):



SG TREND INDEX CONVEXITY VS S&P 500

Key takeaways:

- **Convexity peaks at around an 80-90 day return period.**
- The spike at 160 days is driven by a single big outlier (the 160 days prior to 16/3/2001).
- The broader pattern is for convexity to quickly improve as the return period extends from daily to roughly 80-90 days, then declines again.
- This seems consistent with **Robertson, Goodall/Man AHL (2023)** and highlights the importance of choosing the time frame over which to assess the effectiveness of Trend Following strategies.
- The 90-day period seems long enough for trend following strategies to identify a trend, but too long to respond to any sudden, short-term reversals.

Future Work:

- A better measure of convexity, beyond a simple quadratic line of best-fit coefficient.
- The impact that risk control strategies have on explaining the convexity in addition to the length of the trend signal. Factors such as reducing position size, cutting total risk, and asset selection are likely as important, as discussed by **Tom Wrobel** in recent industry discussions.
- It is genuinely curious how the strategy's convex payoff resembles an option straddle. How being long and short entirely different asset classes, is somehow equivalent being long a put & call on just equities is intriguing.